

Assignment: Multi-Switch VLAN & Inter-VLAN Routing

1.

On Cisco Packet Tracer, create a network with three switches and assign at least two different VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support'). Connect two PCs to each switch and assign each PC to the correct VLAN based on department.

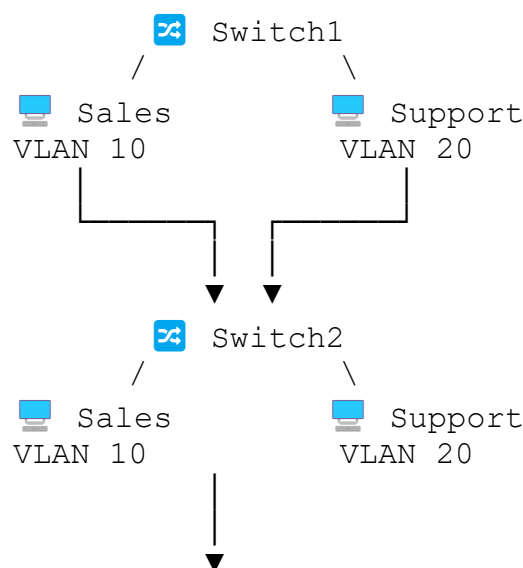
Answer

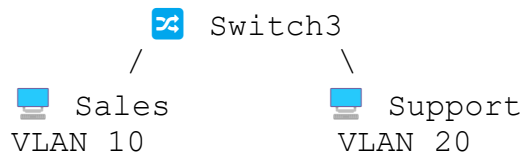
I created a network in **Cisco Packet Tracer** using **three switches** and **six PCs**. I created two VLANs to represent two departments:

-  **VLAN 10 → Sales**
-  **VLAN 20 → Support**

Each switch was connected to two PCs, with one PC assigned to VLAN 10 and the other assigned to VLAN 20.

Network Layout





Create VLANs on Each Switch

I created VLAN 10 and VLAN 20 on all three switches.

```

Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name Support
Switch(config-vlan)# exit
  
```

PC Port Assignment

For example, on each switch:

```

interface fa0/1
  switchport mode access
  switchport access vlan 10
  exit

interface fa0/2
  switchport mode access
  switchport access vlan 20
  exit
  
```

Department Assignment

Switch Port Device VLAN Department

Switch1 Fa0/1 PC1	10 Sales
Switch1 Fa0/2 PC2	20 Support
Switch2 Fa0/1 PC3	10 Sales
Switch2 Fa0/2 PC4	20 Support
Switch3 Fa0/1 PC5	10 Sales
Switch3 Fa0/2 PC6	20 Support

Verification

```
show vlan brief
```

Observation: Each PC was placed into the correct departmental VLAN.

2.

Configure trunk ports between the switches to allow VLAN traffic to pass through. Verify trunking by using the show interfaces trunk command and submit a screenshot of the output.

Answer

I configured the links connecting the three switches as **trunk ports**. Trunking is important because VLAN 10 and VLAN 20 traffic needs to travel between different switches.

Switch1 → Switch2

```
Switch1(config)# interface fa0/24
Switch1(config-if)# switchport mode trunk
Switch1(config-if)# no shutdown
Switch1(config-if)# exit
```

I configured the connected port on Switch2 in the same way.

Switch2 → Switch3

I also configured the ports connecting Switch2 and Switch3 as trunk ports:

```
Switch2(config)# interface fa0/23
Switch2(config-if)# switchport mode trunk
Switch2(config-if)# exit
```

```
Switch3(config)# interface fa0/24
Switch3(config-if)# switchport mode trunk
Switch3(config-if)# exit
```

Verification

On each switch, I used:

```
show interfaces trunk
```

Expected Output

Port	Mode	Encapsulation	Status
Fa0/23	on	802.1q	trunking
Fa0/24	on	802.1q	trunking

Observation: The switch links were working as trunk links, allowing traffic from **VLAN 10 and VLAN 20** to travel between the switches.

 **Screenshot:** I would take a screenshot of the `show interfaces trunk` output from the Packet Tracer CLI and attach it to the assignment.

 **Simple idea:** Access ports connect PCs to a VLAN, while trunk ports carry multiple VLANs between switches.

3.

Set up a Layer 3 device (Router-on-a-Stick using a router or a multilayer switch) to enable inter-VLAN routing between VLAN 10 and VLAN 20. Assign appropriate sub-interfaces and IP addresses for each VLAN.

Answer

I used a **router-on-a-stick** setup to allow communication between VLAN 10 and VLAN 20.

The switch port connected to the router was configured as a trunk.

Switch Configuration

```
Switch(config)# interface gigabitEthernet 0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# no shutdown
```

```
Switch(config-if)# exit
```

Router Configuration

First, I enabled the physical router interface:

```
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# no shutdown
Router(config-if)# exit
```

VLAN 10 — Sales

```
Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit
```

VLAN 20 — Support

```
Router(config)# interface gigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
```

PC IP Configuration

Department Example PC IP Address Default Gateway

	Sales	PC1	192.168.10.10	192.168.10.1
	Support	PC2	192.168.20.10	192.168.20.1

The other PCs in each VLAN can use different IP addresses from the same subnet.

Verification

```
show ip interface brief
```

Observation: The router now has one sub-interface for each VLAN and can route packets between the Sales and Support networks.

4.

Test connectivity by pinging between a PC in VLAN 10 and a PC in VLAN 20. If pings fail, troubleshoot and fix the configuration so that inter-VLAN communication works.
Hint: Check sub-interface IPs, trunk configuration, and default gateways on PCs.

Answer

I tested communication between a **Sales PC in VLAN 10** and a **Support PC in VLAN 20** using the `ping` command.

From the Sales PC:

```
ping 192.168.20.10
```

Initial Result

The first test failed, so I checked the configuration step-by-step.

Troubleshooting Checklist

1. Check VLAN assignment

```
show vlan brief
```

I checked that the Sales PC was in **VLAN 10** and the Support PC was in **VLAN 20**.

2. Check trunk ports

```
show interfaces trunk
```

I confirmed that the links between the switches and the router were configured as trunks.

3. Check router sub-interfaces

```
show ip interface brief
```

I checked that:

```
G0/0.10 → 192.168.10.1  
G0/0.20 → 192.168.20.1
```

were configured correctly and were up.

4. Check PC default gateways

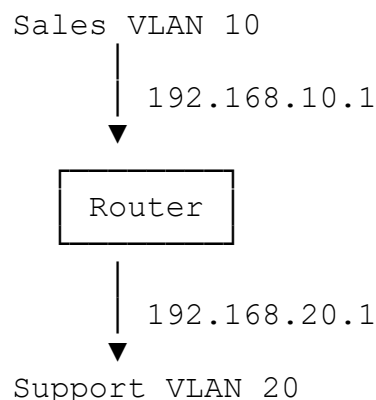
```
Sales PC → 192.168.10.1  
Support PC → 192.168.20.1
```

Fix

After correcting the incorrect configuration, I tested the ping again:

```
PC> ping 192.168.20.10
```

Final Result



Ping Result: Successful 

Final Conclusion

This practical helped me understand how **VLANs, trunk links, and inter-VLAN routing work together**. VLANs separate the Sales and Support departments, while trunk links carry both VLANs between switches. The router-on-a-stick configuration then allows devices from different VLANs to communicate. I also learned that when a ping fails, checking the **VLAN assignment, trunk ports, router sub-interfaces, and default gateways** helps find the problem quickly.

💡 **My understanding:** VLANs create separate departments, trunk links carry their traffic across switches, and the router acts as the **bridge between those departments** when communication is required.